

Classification of Motor Imagery Tasks Using Phase Synchronization Analysis of EEG Based on Multivariate Empirical Mode Decomposition

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Abstract—Phase synchronization has been employed to study brain networks and connectivity patterns. The phase locking value (PLV) is one of the most effective measures widely used for phase synchronization analysis. We first calculate the PLVs of the pair-wise intrinsic mode functions (IMFs) based on multivariate empirical mode decomposition (MEMD) method. Next, the average PLV of the prominent pairs relative to the rest duration is adopted for the classification of motor imagery (MI) tasks. Comparative analysis with the EMD-based PLV method, the proposed method has a significant increase in feature separability for most subjects. This paper demonstrates that MEMD-based PLV method can provide an effective feature in the MI task classification and the potential for BCI applications.

Index Terms—Electroencephalogram (EEG), motor imagery (MI), multivariate empirical mode decomposition (MEMD), phase synchronization, brain connectivity.

I. INTRODUCTION

Human cognition arises from the coordination of anatomically disparate neuronal populations in functionally specialized brain regions [1]. Generally, synchronization of neuronal oscillations associated with different mental tasks can be observed in different functional brain networks. For analysis of brain connectivity between different brain regions, neuronal activity can be measured by synchronization of non-invasively electroencephalographic (EEG) signal.

Many different measures for evaluating the performance of synchronization have been proposed, such as coherence, cross-correlation, phase locking value (PLV), etc [2]. Moreover, a number of studies have suggested that phase synchronization in brain networks may coordinate interaction across anatomically distributed neuronal processing [3], [4]. Therefore, phase synchronization has been widely employed to study brain networks and connectivity patterns.

In contrast to other measures, PLV is used for quantifying the synchrony of two signals by the instantaneous phase difference extracted by using time-frequency transformation.

Additionally, the method of PLV separates signal's phase component from amplitude component and thus may be more suitable to evaluate EEG signal of which the instantaneous amplitude variation comes from eye activity artifacts or other factors. For this reason, PLV is available as the feature of mental tasks such as motor-imagery (MI). However, the limitation of this measure is introduced by using time-frequency transformation to inappropriately decompose EEG signal into a set of narrow-band signals, since the fundamental property EEG signal is almost inevitably nonlinear and non-stationary.

To solve this problem, the empirical mode decomposition (EMD) which decompose the signal into a collection of intrinsic mode functions (IMFs) was utilized to calculate the PLV [5]. There is still a problem with the proposed method, especially when considering that different channel signals may produce the different number of IMFs by the data-driven EMD, and thus IMFs of the same index at multichannel signals span totally different sub-bands. Therefore, when multichannel EEG signals are used to analyze brain connectivity at the same time, it is reasonable to utilize cross-channel information and extract a consistent pattern of multichannel IMFs.

To extend the analysis to multivariate extensions of EMD-based PLV approach, we here propose to evaluate the PLV with the multivariate empirical mode decomposition (MEMD) method. In this paper, the whole MEMD-based PLV analysis of the required multichannel EEG signals is composed of three steps. We first calculate the PLV of signals with MEMD. Next we evaluate the brain connectivity by those PLV results of different EEG channel pairs. Finally we analyze the potential of the PLV of most significant pairs in MI tasks detection.

II. METHODS

A. Materials: Motor Imagery EEG Dataset

The BCI competition IV data set 2a is used for this study [6]. The data of 9 subjects are recorded from twenty-

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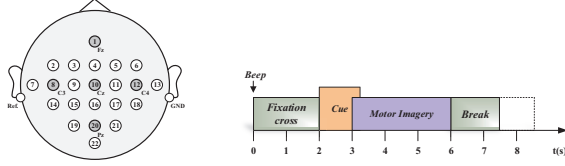


Fig. 1. Electrode montage corresponding to the international 10-20 system (left) and experimental paradigm used in the BCI competition IV data set (right).

two Ag/AgCl electrodes sampled at 250Hz and bandpass filtered between 0.5 and 100Hz . The montage is shown in Fig.1. Each subject performed four MI tasks (left hand, right hand, foot and tongue) according to experimental paradigm (see Fig.1, right). At the beginning, a fixation cross appeared on the black screen with a short acoustic warning tone. After 2s , a cue in the form of an arrow appeared for prompting the subjects to carry out the desired MI tasks until the fixation cross disappeared from the screen at $t = 6\text{s}$. A short break followed where the screen was black again. In addition, the former study revealed that the most reactive frequency components of hand MI varied between 9 and 30Hz , and thus the EEG data is further bandpass filtered before performing other analysis. For illustration, we randomly select the data of 6 subjects ($\#1$ - $\#3$ and $\#7$ - $\#9$), and the analysis of two tasks are presented.

B. Multichannel EEG Signals Decomposition Using MEMD

The MEMD, recently introduced by Rehman and Mandic, is a generic extension of the standard EMD [7]. The main contribution of MEMD method is to compute the local mean of n -dimensional signals by averaging all envelopes which are generated by taking signal projections along different directions, and treat the problem of finding a suitable set of direction vectors as that of finding a uniform sampling scheme on an sphere.

The details of MEMD algorithm for n -dimensional signals $\{\mathbf{x}(t)\}_{t=1}^T = \{x_1(t), x_2(t), \dots, x_n(t)\}$ can be summarized as follows:

- 1) Select a point set for sampling on an $(n - 1)$ sphere.
- 2) Calculate a projection $\{p^{\theta_k}(t)\}_{t=1}^T$ of the original signal $\{\mathbf{x}(t)\}_{t=1}^T$ along the direction vector $\mathbf{d}^{\theta_k}$, giving θ_k as the direction angles on the $(n - 1)$ sphere. For all k direction vectors, then calculate the set of projections $\{p^{\theta_k}(t)\}_{k=1}^K$.
- 3) Identify the time instants $t_j^{\theta_k}$ corresponding to the maxima of the set of projected signals $\{p^{\theta_k}(t)\}_{k=1}^K$.
- 4) Obtain multivariate envelopes curves $\{\mathbf{e}^{\theta_k}(t)\}_{k=1}^K$ by interpolating $[t_j^{\theta_k}, \mathbf{x}(t_j^{\theta_k})]$.
- 5) Calculate the local mean $\mathbf{m}(t) = 1/K \sum_{k=1}^K \mathbf{e}^{\theta_k}(t)$ of the envelopes curves for a set of K direction vectors.
- 6) Subtract the local mean from $\mathbf{x}(t)$, $\mathbf{c}_i(t) = \mathbf{x}(t) - \mathbf{m}(t)$ (i is an order of IMF). Test if $\mathbf{c}_i(t)$ fulfills the stoppage

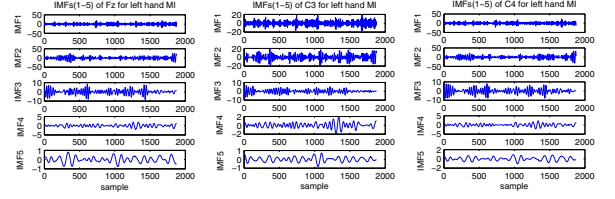


Fig. 2. The IMFs (1-5) of the three EEG channels (C3, C4 and Fz) for left hand MI: subject $\#8$.

criterion for a multivariate IMF. if yes, repeat the above procedure to $\mathbf{x}(t) - \mathbf{c}_i(t)$, otherwise apply the procedure to $\mathbf{c}_i(t)$.

At the end of the decomposition, the original signal $\{\mathbf{x}(t)\}_{t=1}^T$ can be represented as

$$\mathbf{x}(t) = \sum_{i=1}^M \mathbf{c}_i(t) + \mathbf{r}(t), \quad (1)$$

where M is the number of IMFs, $\mathbf{c}_i(t)$ is the i th IMF, and $\mathbf{r}(t)$ is the remaining residue. We here present the EEG data of subject $\#8$ for example. The IMFs of the three EEG channels (C3, C4 and Fz) in left hand MI are shown in Fig.2.

C. MEMD-based Phase Synchronization Detection

To detect the phase synchrony, the MEMD approach is applied to simultaneously decompose multichannel EEG signals into a consistent number of IMFs, denoted by $\{\mathbf{c}_i(t)\}_{i=1}^M$. For each i th IMF, we apply the Hilbert transform [3], which is defined as

$$\hat{\mathbf{c}}_i(t) = 1/\pi PV \int_{-\infty}^{+\infty} \mathbf{c}_i(\tau)/(t - \tau) d\tau. \quad (2)$$

In (2), $\hat{\mathbf{c}}_i(t)$ is the Hilbert transform of the i th IMF, and PV denote the Cauchy principal value. The instantaneous phase of the i th IMF can then be calculated as

$$\Phi_i(t) = \arctan(\hat{\mathbf{c}}_i(t)/\mathbf{c}_i(t)). \quad (3)$$

Under the condition of single-trial, phase synchronization measure PLV between a pair of EEG electrode $\{p, q\}$ in the i th IMF can be defined as:

$$PLV_i(p, q) = 1/T \left| \sum_{t=1}^T \exp(j\{\Phi_i^p(t) - \Phi_i^q(t)\}) \right|. \quad (4)$$

Here, $\Phi_i^k(t)$ is the instantaneous phase of electrode k in the i th IMF level at time instant t . The results of $PLV_i(p, q)$ range from 0 (no phase synchronization at all) to 1 (completive phase synchronization).

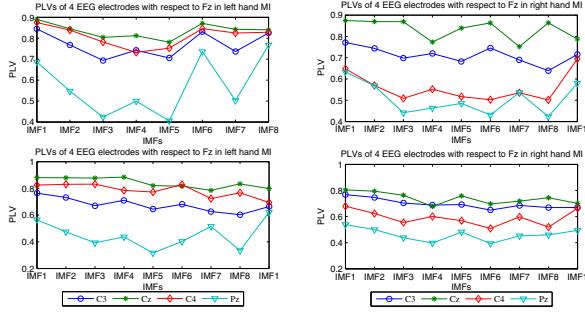


Fig. 3. The PLVs of each IMF of 4 EEG electrodes with respect to Fz during the left and the right hand MI for subject #8. Two different trials of the EEG dataset are illustrated in different panels.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

A. The PLVs of IMFs

The phase synchronization between a pair of EEG electrode $\{p, q\}$ from 1th IMF to M th IMF can be acquired from (4), denoted as $\{PLV_i(p, q)\}_{i=1}^M$. From the results of such PLV set, we estimate the PLVs of four electrodes (C3, Cz, C4, and Pz) with respect to electrode Fz in the IMF level, as shown in Fig.3. The results reveal the similar contralateral dominance as ERD that higher phase synchronization is shown between C4 and Fz for left hand MI while between C3 and Fz for right hand MI.

B. The Prominent Brain Connectivity

The maximum PLV of pair-wise channel signals in the given time window is extracted from $\{PLV_i(p, q)\}_{i=1}^M$. Therefore, the maximum PLV can be defined as,

$$\widehat{PLV}(p, q) = \max\{PLV_i(p, q)\}_{i=1}^M, \quad (5)$$

where M is the number of IMFs. The maximum PLV matrix of the four mental tasks can be constructed from all pair-wise channel combination, denoted by $\widehat{PLV}_{class}(p, q)$ and $class \in \{l, r\}$. Particularly, we consider that the extraction of the maximum PLV may provide us another automatic approach for analyzing subject-specific reactive band.

In order to identify the prominent connectivity from all pair-wise maximum PLV, the variation of PLV relative to the rest duration for the different class of MI tasks can be calculated by

$$\Delta PLV(p, q) = \frac{\widehat{PLV}_{class}(p, q) - \widehat{PLV}_{rest}(p, q)}{\widehat{PLV}_{rest}(p, q)}. \quad (6)$$

As illustrated in Fig.4, the more prominent pair-wise connectivity for both MI tasks corresponds to higher ΔPLV , most of which concentrate in one row or one column.

Then, taking the #8 subject for example, the mapping of brain connectivity for both left and right hand MI tasks at the 3 – 6s epoch is shown in Fig.5. The red shadowed electrodes are the ones corresponding to the most prominent

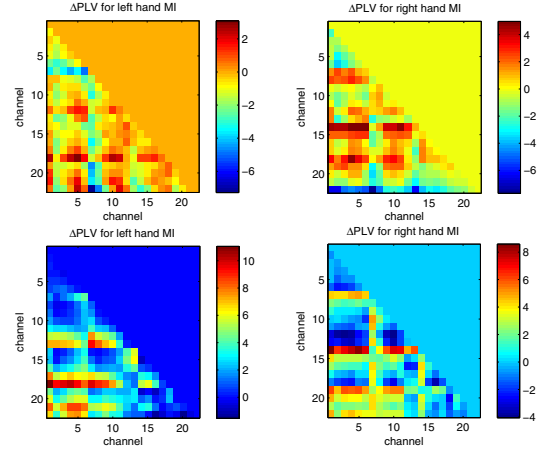


Fig. 4. All possibly pair-wise connectivity of left and right hand MI tasks for subject #1 (top) and subject #8 (bottom).

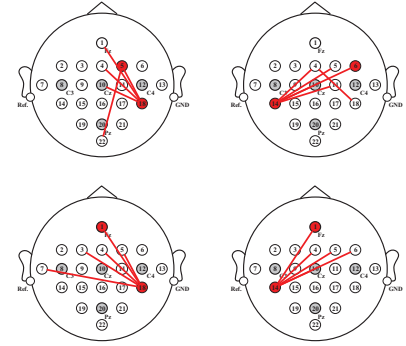


Fig. 5. The EEG connectivity mapping of left and right hand MI tasks for subject #1 (top) and subject #8 (bottom), and the corresponding figure of the both hand tasks is from left to right.

connectivity. The 5 most prominent pair-wise electrodes for both MI tasks are selected, and the majority appears in their contralateral region of brain.

C. The Comparing Results of Two Methods

According to Table I, the PLVs of the 5 most prominent pair-wise electrodes over time for both tasks are first selected, as illustrated in Fig.6. Here, the 0.3s time window is chosen to calculate PLVs using (5) as $P(n)$, where n represents each epoch. Then the average PLVs $\bar{P}(n)$ of these 5 prominent pairs for different MI tasks are calculated. The results of these PLVs are obviously increased on the pair-wise electrodes of the contralateral region corresponding to both hand tasks during MI period (3 – 6s), while their ipsilateral areas mainly have no significant change. Therefore, these prominent pair-wise electrodes on the two sides of the brain cortex region can be used for distinguishing the left or right hand MI tasks. For classification, PLV over the rest duration is represented

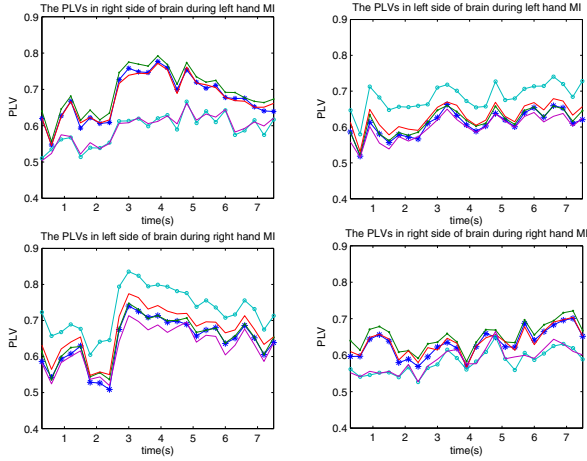


Fig. 6. The PLVs of the 5 most prominent pair-wise electrodes in different side of the brain for both left and right hand MI tasks: subject #8.

by R, and these average PLVs relative to the rest duration as:

$$P_r(n) = (\bar{P}(n) - R)/R * 100, \quad (7)$$

where $P_r(n)$ of the selected pair-wise electrodes can be adopted to classify left or right hand MI. As illustrated in Fig.7, the results of EMD-based PLV are compared with these obtained from the method based MEMD, which makes it easier to distinguish different tasks.

Fisher score is then utilized to measure the separability of the mentioned features for left and right hand MI, illustrated as FS_EMD and FS_MEMD for the two methods, and

$$\Delta FS(\%) = (FS_EMD - FS_MEMD)/FS_EMD * 100. \quad (8)$$

The comparing results between EMD-based PLV and MEMD-based PLV are shown in Table I. This proposed method based on MEMD has a significant increase in feature separability for most subjects. The increase in Fisher score can identify the improvement with the MEMD-based PLV method, and an average increase of 104.5% is obtained.

TABLE I

THE COMPARING RESULTS OF FISHER SCORE FOR THE TWO METHODS

	S#1	S#2	S#3	S#7	S#8	S#9
FS_EMD	0.0464	0.0348	0.7391	0.0073	1.6958	0.2560
FS_MEMD	0.2247	0.0552	1.9502	0.0096	2.0553	0.1724
$\Delta FS(\%)$	384.3	58.6	163.9	481.8	21.2	-32.7

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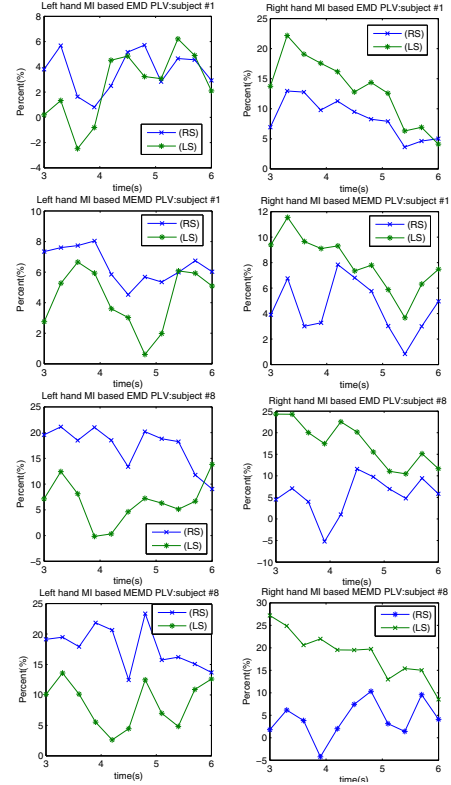


Fig. 7. Calculation of the average PLVs relative to the rest duration in left side (LS) and right side (RS) of the brain during the two kinds of tasks, and comparison of the results of two methods: subject #1 and subject #8.

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